Statistical instability of a stochastically forced shear

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Abstract

In this article, we prove that the stochastic Navier–Stokes equation driven by shear noise admits, in an appropriate scaling regime (of small aspect ration and small viscosity), two invariant measures.

1 Setting and main results

Consider the 2D stochastic, incompressible Navier–Stokes equations

$$\partial_t u + u \cdot \nabla u = \nu \Delta u + \xi + \nabla p, \qquad \operatorname{div}(u) = 0,$$

driven by noise ξ white in time and smooth in space.